

Name: _____

Date: _____



EARLY ALGEBRA



LEVEL 1: MISSING NUMBERS

1 Fill in the missing number. ★

1 $3 + \square = 7$

5 $2 \times \square = 8$

2 $\square + 5 = 9$

6 $\square \times 3 = 12$

3 $10 - \square = 6$

7 $\square \div 2 = 3$

4 $\square - 2 = 5$

8 $16 \div \square = 4$

2 Find the rule. ★

Write the missing number(s) in the box(es).

1 2, 4, $\square$, 8, 10 Rule: + 2

2 15, 12, $\square$, 6, 3 Rule: - 3

3 3, 6, 9, $\square$, 15 Rule: + 3

4 20, 18, $\square$, $\square$, 12 Rule: - 2

3 Balance the boxes. ★

Write the missing number in each box to make the number sentence true.



$\square + 2 = 6$



$9 = \square + 4$



$\square - 3 = 5$



$7 = \square - 2$

4 Challenge! ★

Think carefully and fill in the missing number(s).

1 $\square + \square = 10$

Clue: One of the numbers is 4.

What are the two numbers?

_____ and _____.

2 $14 - \square = 9$

What is the missing number?

_____.



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LEVEL 3: SYMBOLS

1 Find the value of each symbol.

Use the equations to find the value of each shape.

$$\square + \square = 10 \quad \square = \square$$

$$\bigcirc + \bigcirc + \bigcirc = 15 \quad \bigcirc = \square$$

$$\triangle + \bigcirc = 9 \quad \triangle = \square$$

$$\star + \triangle = 11 \quad \star = \square$$

2 Fill in the missing numbers.

Replace each symbol with the correct number to make the equation true.

$$\heartsuit + 4 = 10 \quad \heartsuit = \square$$

$$\diamondsuit - 3 = 5 \quad \diamondsuit = \square$$

$$\bullet + 7 = 13 \quad \bullet = \square$$

$$\blacktriangle + 6 = 12 \quad \blacktriangle = \square$$

3 Picture equations.

Write a number sentence for each picture using a symbol, then solve it.

$$\text{apple} + \text{apple} + \text{apple} = 12$$

Let $\text{apple} = \square$ $\square + \square = \square$

$$\star + \star + \star + \star = 16$$

Let $\star = \square$ $\square + \square + \square = \square$

$$\text{butterfly} + \text{butterfly} + \text{butterfly} - \text{butterfly} = 8$$

Let $\text{butterfly} = \square$ $\square - \square = \square$

$$\text{flower} + \text{flower} + \text{flower} = 20$$

Let $\text{flower} = \square$ $\square + \square = \square$

4 Match the expressions.

Draw a line to match each expression on the left with an equivalent expression on the right.

A. $\diamondsuit + 5$

1. $6 + \blacktriangle$

B. $7 + \bullet$

2. $3 + \star$

C. $\blacktriangle + 6$

3. $\bullet + 7$

D. $\star + 3$

4. $5 + \diamondsuit$

★ CHALLENGE! ★

If $\star = 2$, $\bullet = 5$ and $\blacktriangle = 8$,
what is the value of $\star + \bullet + \blacktriangle$?



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LEVEL 4: EQUATIONS



1 Solve the equations.

Find the value of x .

$$1 \quad x + 3 = 7 \quad x = \square$$

$$2 \quad 9 - x = 4 \quad x = \square$$

$$3 \quad 2 \times x = 10 \quad x = \square$$

$$4 \quad x \div 2 = 4 \quad x = \square$$

$$5 \quad x + 6 = 11 \quad x = \square$$

$$6 \quad 15 - x = 8 \quad x = \square$$

2 Match each equation to its answer.

Draw a line to match.

$$A \quad x + 5 = 12 \quad \bullet \quad \bullet \quad 1 \quad x = 4$$

$$B \quad 14 - x = 6 \quad \bullet \quad \bullet \quad 2 \quad x = 7$$

$$C \quad 3 \times x = 18 \quad \bullet \quad \bullet \quad 3 \quad x = 6$$

$$D \quad x \div 3 = 5 \quad \bullet \quad \bullet \quad 4 \quad x = 8$$

$$E \quad 2 \times x = 8 \quad \bullet \quad \bullet \quad 5 \quad x = 15$$

3 Complete the equations.

Fill in the missing number to make each equation true.

$$1 \quad x + \square = 9$$

$$2 \quad 12 - x = \square$$

$$3 \quad 4 \times x = \square$$

$$4 \quad x \div 5 = \square$$

$$5 \quad \square + 7 = 13$$

$$6 \quad 18 - \square = 10$$

Think carefully!



4 Challenge!

Find the value of x and check your answer by balancing both sides.

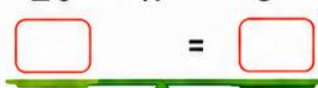
$$1 \quad x + 7 = 15$$



$$2 \quad 2 \times x = 18$$



$$3 \quad 20 - x = 5$$



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LEVEL 5: SIMPLE FORMULAS

1 Use the formula!



Use the formula to fill in the table.



Formula:

Number of wheels = $2 \times$ number of bikes

Number of bikes	1	3	5	7	10
Number of wheels					

How many wheels are there if there are 9 bikes?

2 Use the rule!



Use the rule to complete the table.



Rule:

Legs = $4 \times$ number of dogs

Number of dogs	2	4	6	8	11
Total legs					

How many legs are there if there are 12 dogs?

3 Use the formula!



Use the formula to solve each problem.



Formula:

Stickers = $3 \times$ packs

1 How many stickers in 1 pack?

2 How many stickers in 4 packs?

3 How many stickers in 7 packs?

4 How many packs are needed for 24 stickers?

4 Challenge!



Use the rule to find the missing outputs.

	Input (n)	Rule	Output
1	5	Output = $5 \times n$	
2	9	Output = $3 \times n + 1$	
3	7	Output = $2 \times n + 4$	

4 Think of your own rule.
Use $n = 6$.

My rule: _____

Output when $n = 6$:

